

Dissatisfied Creators: Generating Creative Ideas Amid Negative Emotion in Health Care

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Abstract

Improving performance often requires creative ideas in extreme work settings like health care. Frontline workers are promising sources of creative ideas. The authors assert that their job dissatisfaction is positively associated with creativity in health care settings because negative emotion spurs creativity when tied to engaging work and that characteristics such as shorter tenure, greater role centrality, and high boundary spanning can strengthen this relationship. The authors find supporting evidence in data on ideas generated over 18 months by health professionals in 12 clinics.

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Results suggest that health care organizations should not overlook dissatisfied workers as valuable sources of creative improvement ideas.

Keywords

creativity, health care, emotions, dissatisfaction, engagement

Creative ideas are needed to improve health care delivery, that is, the work processes used by healthcare organizations and professionals to prevent, diagnose, and treat illness and preserve patient health (Institute of Medicine, 2002). Despite efforts to improve over several decades, industry statistics indicate poor performance on many clinical quality and experiential metrics, and high cost for patients, raising questions about the value of care received by patients in the United States (Porter, 2010). Compounding this situation, the U.S. healthcare industry also faces new population and policy demands (Dzau, McClellan, McGinnis, et al., 2017). A growing and older patient population with more complex and chronic care needs is rendering current processes inadequate (Coleman, Austin, Brach, & Wagner, 2009). Creative ideas, that is, ideas that are novel and useful, are necessary in environments characterized by change and complexity (Amabile, 1988). Hence, there is increasing interest in creativity, the generation of creative ideas, in health care.

A key issue for health care, like other industries, is who can best generate creative ideas (Shalley, Zhou, & Oldham, 2004). Health care professionals, for example, physicians and nurses, may be promising sources of creative ideas given their familiarity with current practices, motivation to provide the best possible care for patients, and frequent generation of work-arounds to problems (Axtell, Holman, Unsworth, Wall, & Waterson, 2000). However, health care professionals also face significant structural barriers to creativity, for example, limited time for reflection and experimentation due to high workloads, organizational cultures that stifle ideas that depart from current practice, an industry emphasis on evidence-based medicine and risk minimization, and a professional hierarchy that privileges creativity by higher status individuals and discourages the initiative of lower status individuals (Edmondson, 1999). These challenges can make creativity professionally and interpersonally risky in health care and depart from the

experience of workers in artistic communities (e.g., musicians) where individuals are socialized toward persistent creativity and normalizing failure (Mueller, Melwani, & Goncalo, 2012; Skaggs, 2019). Personal factors are barriers to creativity too. Health care professionals report among the highest levels of burnout and job dissatisfaction of any professional group (Dyrbye & Shanafelt, 2016). More than half of U. S. physicians, nurses, and administrators report burnout from and dissatisfaction with their jobs (Shanafelt et al., 2015). Job dissatisfaction refers to a low level of contentment with one's job (Farrell, 1983). It is an emotional state that can precipitate emotional decoupling of a person's sense of self from their work role (Farrell, 1983) and can result in a withdrawal of commitment and effort, or exit from the workplace (Brockerman & Ilmakunnas, 2008). In a setting like health care with these challenges, it would be logical to assume limited creativity, especially by those who are dissatisfied.

We, however, assert that job dissatisfaction, although generally an undesirable, negative emotional state, is positively associated with creativity in health care. Prior research typically shows that positive emotions are associated with creativity (Amabile, Barsade, Mueller, & Staw, 2005; Isen, Daubman, & Nowicki, 1987). Conversely, meta-analyses of the limited work on negative emotions and creativity suggest a negative relationship (Baas, De Dreu, & Nijstad, 2008). A few studies however intimate that negative emotions can be positive for creativity if they foster persistence in problem-solving and cognitive manipulation (De Dreu, Baas, & Nijstad, 2008; Martin, 2001). Those few studies have focused on negative emotions unrelated to work experience (e.g., anger and disgust), leaving open the question of how negative emotions related to work affects creativity at work. Researchers have noted that, "especially when it comes to negatively toned mood states, more research is badly needed" (De Dreu et al., 2012, p. 223).

In this article, we focus on job dissatisfaction because of its prevalence in health care and other industries and the severity of its potential consequences for workers and organizations (e.g., decoupling of sense of self and turnover costs). We posit that job dissatisfaction, though a negative emotion, can be positively associated with creativity when it is tied to activating and engaging work. Health care delivery is an epitome of engaging work, involving an aggregate of unique and extreme work challenges. In the next section, we identify unique and extreme work challenges in health care. We then present the argument for job dissatisfaction being a catalyst for creativity in such extreme work contexts due to the persistence generated by these circumstances. We

argue that this relationship may be especially strong for individuals with particular job attributes as attributes shape the experience of work and exposure to information that might inform creativity. As such, we also examine person–job characteristics likely to strengthen this negative emotion–positive cognition relationship. We test our hypotheses about job dissatisfaction and moderating person–job characteristics using data on improvement ideas generated by individuals working in 12 clinics over an 18-month period.

The Engaging, Unique, and Extreme Work of Health Care Delivery

Health care delivery is an extreme work context characterized by the unique work condition of risk of patient death as a work outcome. Health care is also extreme in having a high level of multiple work conditions relative to the norm including high ambiguity due to variable patient response to treatment and high emotions due to patient suffering (Driskell, Salas, & Driskell, 2018). Despite meeting the definition of an extreme work context on each criterion (Gilmartin & D'Aunno, 2007), it aspires to high reliability, that is, to operate nearly error-free amid complexity, interdependence, and time pressure (Vogus & Sutcliffe, 2007). High-reliability work demands performing at peak levels at all times, so lives are not lost. In health care, these stakes are compounded by high levels of ambiguity. Complexity and idiosyncrasies of patients, the uncertain pathways that their diseases may follow, and dynamism in clinical knowledge introduce ambiguity that complicates diagnosis and treatment (Nembhard et al., 2006).

These challenges demand distinct modes of working. Workers at the frontlines coproduce care (service) experiences with patients, reflecting the reality that workers possess clinical expertise and patients possess self-expertise that must be integrated for high-quality care (Vogus & McClelland, 2016). Integration requires ambidexterity by workers who must balance high levels of customization to ensure patient-centered care with standardized yet evolving, evidence-based practice to prevent harm (Bohmer, 2005). This work often takes place in the midst of high emotion as patients and workers contend with suffering, uncertainty, and vulnerability related to care. The intensity and emotionality of coproduction necessitates high attention, focus, and commitment (Vogus & McClelland, 2016). This responsibility is frequently shared by many individuals who form a patient's care team. Each brings specialized knowledge, which creates the demand for effective teamwork to

integrate their knowledge. Difficulty meeting this challenge is cited as a key reason for the prevalence of poor-quality, high-cost care (Institute of Medicine, 2012).

In addition, health care delivery is distinct and extreme in that it holds great meaning for many of its workers, as the goal of this work is to improve health and save lives (West, Dyrbye, Erwin, & Shanafelt, 2016). Many view their work as a calling, a description that has its roots in the philosophical origins of health care as a healing profession (Dyrbye & Shanafelt, 2016). While professionals in other industries often feel called to serve their customers, health care workers arguably face the highest standard to realize their calling: Have I fully worked to save lives, improve health, and do no harm? Together, these features of health care constitute extreme work conditions and create unique challenges for workers in terms of emotions and creativity.

Dissatisfaction and Creativity in Extreme Work Conditions: Background and Hypotheses

Scholars have produced a body of research on the effects of positive and negative emotions on creativity (Baas et al., 2008). Positive emotions such as joy are theorized to prompt individuals to broaden-and-build by discarding routines and pursuing novel actions (Fredrickson, 2001). This theory is supported by Isen et al.'s (1987) laboratory studies, which showed that positive mood states are associated with more creative word associations. Outside of the laboratory, with employees from companies where individuals do creative thinking to produce novel consumer products, Amabile et al. (2005) found that self-rated positive mood was associated with more creative thoughts. These representative studies have been affirmed by a meta-analysis that shows a robust, positive link between positive emotions and creativity, particularly for emotions that are activating, associated with approach motivation (moving toward desired outcomes), and promotion-focused such as happiness (Baas et al., 2008).

In contrast, studies of negative emotions indicate heterogeneous effects with respect to creativity (Baas et al., 2008). The dual pathway to creativity model (DPCM; De Dreu et al., 2008) explains these effects as well as the observation that positive and negative moods can generate creativity by (a) distinguishing between activating and deactivating types; (b) specifying that activating moods—both positive (e.g., happy) and negative (e.g., anger)—are more likely to spark creativity than deactivating moods (e.g., feeling relaxed); and (c) noting that situational

variables like moods may influence creativity through different pathways—via cognitive flexibility (flexible processing of information), cognitive persistence (persistent probing, and systematically combining elements and possibilities), or both. Positive emotions activate creativity through cognitive flexibility, whereas negative emotions activate creativity through cognitive persistence (De Dreu, Baas, & Nijstad, 2011).

Satisfaction and dissatisfaction have been neglected in creativity studies, although they are widely studied constructs in organizational behavior (Baas et al., 2008). DPCM theorists hold that these emotions differ from the transient mood states that are their focus (anger or sadness), and so this research stream has purposely excluded them. A study of employee engagement however found that satisfaction, like other positive emotions, is weakly positively associated with creativity (Eldor & Harpaz, 2016). The one study that examined the dissatisfaction–creativity relationship found that dissatisfaction was associated with creativity when organizational support for creativity, coworker support and feedback, and commitment to remain with the organization were high (Zhou & George, 2001). The researchers did not consider a main effect of dissatisfaction because “it is naive to assert that job dissatisfaction will always lead to creativity in the workplace . . .” since job dissatisfaction is known to “lead to turnover which negates the possibility of dissatisfied organizational members trying to improve conditions in the organization through creative performance” (Zhou & George, 2001, p. 682). Consequently, they suggest identifying conditions under which job dissatisfaction fosters creativity.

We assert that a positive job dissatisfaction–creativity relationship exists in healthcare delivery due to the unique and extreme work of coproducing highly reliable care, which is difficult but also activating and engaging. Caring for patients motivates many health care workers to dedicate great effort to care, even those that report dissatisfaction (McNeese-Smith, 1999). In other words, the core work of health care activates workers. The DPCM and related research suggests that feeling activated is critical for creativity; deactivation limits creativity (De Dreu et al., 2008). When there is dissatisfaction, there is unsettling discomfort (Deci, Connell, & Ryan, 1989). We propose that when unsettling discomfort occurs in an activating context, it catalyzes dedicating time and effort to in-depth problem assessment, generating creative ideas, and enabling the cognitive persistence needed for breakthroughs (De Dreu et al., 2008). Research suggests that when individuals are activated and feel negative emotion (like dissatisfaction), they become more analytical and process information in a more bottom-up way (Derryberry & Reed, 1998).

Prior work further indicates that creativity is fostered in organizational contexts that direct attention toward the need for creativity to address problems (i.e., are activating) and maintain individuals' interest in generating new ideas (Shalley et al., 2004). Research on creativity in precarious work circumstances shows that passion for the job can be activating and foster persistence rather than exiting the field in search of greater stability (Umney & Kretsos, 2015). In other less extreme, less activating work settings, dissatisfaction is likely to prompt individuals to exit or passively remain with the organization (Hirschman, 1972) rather than dedicate effort required for creativity and surmounting the barriers to it. Therefore, we proposed the following hypothesis:

Hypothesis 1: Job dissatisfaction is positively associated with idea creativity in extreme work environments such as health care delivery.

Person–Job Characteristics That Moderate the Dissatisfaction–Creativity Relationship

Research suggests that creativity depends not only on task engagement, which is a function of organizational context and, as we argue, is provided in part by dissatisfaction in activating extreme work settings, but also on the individual and his or her circumstances (Eldor & Harpaz, 2016). In work environments less extreme than health care, contextual variables (e.g., supervisor support), task domain-relevant skills (materials drawn on during operation such as mathematical reasoning and drawing), and personality variables (e.g., openness to experience) have been identified as moderators of main effect relationships to creativity—strengthening or attenuating the effects of independent variables on individual creativity (Conti, Coon, & Amabile, 1996; Perry-Smith & Shalley, 2003). Person–job characteristics of individuals such as role have received little attention as potential moderators, despite these characteristics being widely used to describe individuals and define their work across industries (George, 2007). Person–job characteristics are the attributes that characterize a person's circumstances in relation to her or his job. We expect these characteristics to be particularly relevant in the extreme work environments because they shape the experience of work, as well as individuals' exposure to diverse information that might inform their creativity. The diversity is likely to be consequential due to the complexity of extreme work.

We propose three person–job characteristics—organizational tenure, role centrality, and boundary-spanning—not only influence creativity but also moderate the creativity of the dissatisfied individuals in extreme work settings like health care. Creativity is often described as a product of cognitively combining new and nonredundant information together, termed *cognitive flexibility* (De Dreu et al., 2008). Each of the three person–job characteristics likely provides knowledge that enables cognitive flexibility in a distinct way. We propose that this flexibility affects the creativity that arises from dissatisfaction: the greater cognitive flexibility (from knowledge) alongside cognitive persistence (motivation due to dissatisfaction), the greater creativity.

Organizational Tenure

Time affiliated with an organization determines a worker's perspective on the organization and his or her job (Shalley et al., 2004). Shorter tenure is associated with a novelty-seeking perspective; newcomers *search for knowledge* due to lack of familiarity with the organization and how their job operates within it (Ford, 1996). New recruits often display a desire to learn, ask questions, and explore ways to have organizational impact (Ng & Feldman, 2010). They are able to do so due to the privilege of newcomer status (March, 1991; Weick & Roberts, 1993), and as they integrate responses with their prior knowledge, they may generate creative possibilities. Their learning orientation enhances exposure to new and diverse information inputs and facilitates their cognitive flexibility (Simonton, 1999). In other words, the search for knowledge channels and directs the energy of dissatisfaction to translate possibilities into creative ideas (Kunda, 1990). Dissatisfaction may be particularly important in health care because it helps cut through the competing demands on energy and attention (Ford, 1996). For longer tenured individuals, dissatisfaction is likely to generate ideas that may have high levels of usefulness, based on accumulated knowledge of the organization and expertise, but lower levels of novelty due to familiarity with the organization (Baron, Davis-Blake, & Bielby, 1986). In contrast, the discomfort of dissatisfaction coupled with the cognitive flexibility that comes with the fresh, exploratory view of those with shorter tenure is associated with higher creativity.

Hypothesis 2: Dissatisfaction is associated with greater idea creativity for those with shorter organizational tenure than those with longer tenure in extreme work environments like health care delivery.

Role Centrality

An individual's job role determines his or her centrality in work teams, whether he or she is a central or supporting member (Hackman, 1987). In health care, the closer the role is to direct patient care, the more central the role is in the care team (Freidson, 1970). Physicians and nurses are considered central roles, whereas allied professionals (e.g., medical assistants) are considered supporting roles. We propose that those in central roles generate more creative ideas related to core work processes (e.g., improving patient experiences and outcomes) than those in supporting roles because of their *depth of knowledge*, cultivated by their more extensive training and involvement in more aspects of patient care (Linzer et al., 2000). In extreme work settings such as health care, this depth of knowledge confers increased accountability for (patient) outcomes, which can intensify the sense of responsibility to the patient. Those in central roles, as a hallmark of their work, weave together fine-grained, heterogeneous information such as patients' preferences, symptoms, and clinical evidence to diagnose problems and generate solutions, that is, care plans (Linzer et al., 2000). This process of receiving and combining new and diverse information is a hallmark of creative individuals. Thus, we expect creativity to be more impactful for those with role centrality. When they are dissatisfied, we expect that their creativity is enhanced because their drive to develop creative improvement ideas is augmented by the depth of knowledge that helps make their novel ideas also useful and implementable. In contrast, less central individuals lack the depth of knowledge to offer sufficiently novel or useful ideas. Therefore, we expect that

Hypothesis 3: Dissatisfaction is associated with greater idea creativity for those with more central roles than those with less central/supporting roles in extreme work environments like health care delivery.

Boundary Spanning

Boundary spanners, those who through their interactions link their internal networks with external sources of information (Aldrich & Herker, 1977), tend to have social networks with a larger number of ties to individuals outside of their immediate team. In contrast, those who are internally oriented have a larger number of ties within their immediate team (Perry-Smith & Shalley, 2003). In health care, where team-based care models are growing, an internal team orientation frequently dominates patterns of interactions as each care team takes

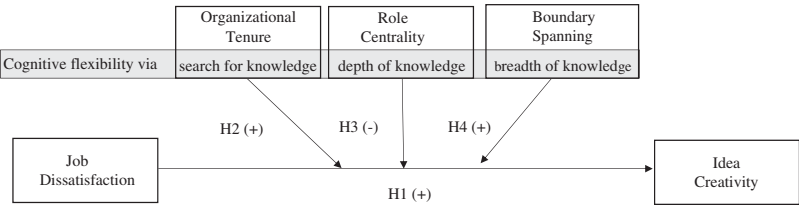


Figure 1. Conceptual framework.

responsibility for a patient panel and focuses attention on developing shared mental models, in which core members have a shared understanding of the task (Bodenheimer, Wagner, & Grumbach, 2002). Recently, however, boundary spanning has been encouraged to foster coordinated care with all those relevant to patients’ health, including other care providers and service organizations (Sikka, Morath, & Leape, 2015). Boundary spanners, similar to those in central roles, necessarily accumulate skills and experience in combining diverse information. Often boundary spanners have the additional need and challenge of combining insights across boundaries, a skill that develops cognitive flexibility and lends itself to creativity. Dissatisfaction may be especially activating for them because their external activity exposes them to different ways of working, which demonstrates the potential benefits of creativity. While dissatisfaction is likely to foster creativity for the internally oriented too, presuming commitment to performance, their internal focus may limit their ability to capitalize on motivation provided by dissatisfaction relative to boundary spanners. Thus:

Hypothesis 4: Dissatisfaction in extreme work environments like healthcare delivery is associated with greater idea creativity by workers who span boundaries more (e.g., interact with others outside their team) than those who span boundaries less.

Together, our hypotheses lead to an emotions-based model of creativity in extreme work environments like health care, presented in Figure 1.

Methods

Study Design and Setting

We tested our hypotheses by performing a prospective panel analysis of 220 improvement ideas generated over 18 months by 72 clinical staff

serving on 12 quality improvement teams in 12 federally qualified community health centers (one team per center). Roles included primary care providers (PCPs; i.e., physician or advanced practitioner), nurses, medical assistants, and behavioral health providers. The centers are part of a single organization in one state in the United States. They provide comprehensive primary care services to more than 130,000 patients a year and have a special commitment to serving the uninsured, underinsured, and special populations (e.g., patients with chronic mental health issues). A key quality improvement strategy was to form improvement teams charged with developing quality-improving innovations for each center. Each team included at least one member from each clinical role, with two per role in larger centers. Teams had six members on average and met monthly. During our study period, the organization tasked the teams with developing and implementing innovations focused on improving care coordination particularly for patients needing chronic disease management or posthospital care because its self-study identified care coordination as an issue for many of these patients.

Data Sources

We used three sources of data on team members and their ideas for this study: improvement team meeting transcripts, a staff work experience survey, and sociometric sensors. Data on center characteristics were obtained from the organization's business intelligence department.

Quality improvement team meeting transcripts. Quality improvement team discussions were documented in 216 meeting transcripts (18 monthly meetings for 12 centers), which we used to identify improvement ideas and their proposers ("the creators"). Transcripts in all centers contained the following standardized information: center name, date, time and location, attendance, agenda topic, deliverable, outcome measure, staff responsible, notes/discussion, and action items (action, owner, and due date). The first author analyzed the transcripts using content analysis techniques (Miles & Huberman, 1994) to identify ideas for improving care coordination and the individual(s) who proposed each idea. Ideas were defined as "a different alternative for a possible course of action to approach the task at hand" pp. 444 (Binnewies, Ohly, & Sonnentag, 2007). A second researcher independently coded ideas in a randomly selected 30% of transcripts as a check on the process. There was 83% agreement between coders on idea identification, which falls

within the intercoder reliability range of 70% to 94% considered “acceptable” to “exceptional” (Campbell, Quincy, Osserman, & Pedersen, 2013). Coders discussed disagreements, which led to refinement of the identification rubric. All ideas were coded again to ensure consistency. In total, we identified 220 improvement ideas in the transcripts. All 72 team members offered ideas.

Staff work experience surveys. We collected information on team member characteristics, for example, role, organizational tenure, and job dissatisfaction, using a survey. We recruited participants using e-mails from center leadership and lunchtime meetings with members of the research team, where consent forms were signed. At the start of the 18-month study period, we administered the survey via e-mail or paper to all 72 team members in the 12 centers, with 100% participation. Participants were full-time employees. The majority were female (79%). The largest group of respondents were nurses (31%), with PCPs (27%), medical assistants (23%), and behavioral health providers (19%) comprising the other groups, and 61% of staff were employed by the organization for more than 2 years.

Sociometric sensors. In the week or two following survey completion, we collected social network data using wearable sensors, which have been used for studies of workplace interactions in hospitals (Isella, Romano, & Barrat, 2011) and nonhealth settings (Waber, 2013). The sensors, made by Humanyze, formerly Sociometric Solutions, Boston, MA were the size of a deck of cards and worn around the neck. They contained sensors that recorded each time an individual spoke with another individual wearing a sensor (not speech content), which allowed us to obtain individuals’ network of interactions inside and outside their assigned care team in a nonobtrusive fashion (Olguin Olguin, Gloor, & Pentland, 2009). The validity and reliability of the data collected by such sensors have been established in other studies (Chaffin et al., 2015).

We used interaction data collected from primary care teams at each center (which included the 72 members of the improvement teams in this study) during 1 week of work (Monday–Saturday). We used one week that the centers indicated was a representative week to capture multiple days of interaction in “a normal week.” This allowed us to capture average interactions versus rely on a potentially outlier day(s). Past research found that network characteristics are relatively stable, adding to the reliability of this multiday, slice-of-time approach (Olguin Olguin et al., 2009). In three centers, the sensor-wearing period

coincided with improvement team meetings (3 of 216 meetings; 3 of about 576 hours in the sensor period). As our interest was individuals' ties broadly, overlap with improvement team meetings was not significant for this work.

Measures

Idea creativity. Ideas identified in meeting transcripts were rated for creativity by an expert panel using the Consensual Assessment Technique, a reliable and valid technique used in organizational studies (Amabile, 1982). In this method, experts familiar with the domain independently rate an idea or object for degree of novelty and usefulness, and these ratings are multiplied to generate a single creativity score. The first author recruited five healthcare executives to serve as experts raters. All worked full-time in healthcare administration roles (including quality improvement functions) in the United States. The majority were female (60%). All had at least 10 years of professional experience in healthcare delivery settings and graduate degrees in healthcare administration. The experts were not aware of the others participating in the assessment. Each expert was e-mailed a survey that contained short vignette descriptions of the ideas from transcripts (maximum three sentences) and was asked to rate each idea ($N=220$) for level of novelty (1 = *least novel* to 5 = *most novel*) and level of usefulness (1 = *least useful* to 5 = *most useful*). Order of ideas was randomized for each rater to prevent order effects.

In accordance with the Consensual Assessment Technique, novelty and usefulness scores for each idea were multiplied together to generate an overall score ranging from 5 to 25. These overall scores were divided by 5 to generate a final creativity score out of 5 (1 = *least creative* to 5 = *most creative*). We averaged across the five experts' scores to generate one average creativity score for each idea. To assess the interexpert reliability of scores, an intraclass correlation (ICC) was calculated. The $ICC_{1,k}$ of .79 indicated a moderately high degree of consistency between experts that is considered acceptable for assessments of creativity (Amabile, 1982).

Job dissatisfaction. Job dissatisfaction for each idea generator/individual was measured using a reverse-scored measure of job satisfaction, assessed by responses to the staff survey question: "Overall I am satisfied as an employee of [organization name]." The response categories ranged from 1 = *Strongly disagree* to 4 = *Strongly agree*. Similar,

Singleton measures of job satisfaction have been used in previous studies testing the link between job satisfaction and other outcome variables (Allen & Van der Velden, 2001). We converted the satisfaction measure into our job dissatisfaction measure, similar to Zhou and George (2001), by subtracting the job satisfaction value from 5, creating a measure ranging from 1 = *low dissatisfaction* to 4 = *high dissatisfaction*. Because past research suggested that a curvilinear relationship might exist between emotions and creativity (James, Brodersen, & Jacob, 2004) we also calculated the quadratic of this measure to assess this possibility.

Role centrality. The health care literature identifies two roles as most central in the primary care team—nurses and PCPs, which includes physicians and advanced practitioners—and all other roles as supporting these (Hall, 2005). We obtained each individual's role from response to the staff survey item: "Please select the job title which best describes your position." This item had four response categories: PCP, nurse, behavioral health provider, and medical assistant. We created a binary variable for role centrality equal to 1 for PCPs and nurses and equal to 0 for medical assistants and behavioral health providers.

Organizational tenure. We obtained the organizational tenure of each individual using responses to the staff survey question: "How long have you been employed by [organization name]?" Response categories ranged from 1 = *Less than 6 months* to 6 = *10 or more years*.

Boundary spanning. Boundary spanning of each individual was measured using the external to internal index (E-I Index), an established social network measure (Krackhardt & Stern, 1988), which was calculated using data from the sociometric sensors. This index is a measure of a person's relative density of ties inside a social group to ties outside of the group. Assuming two groups based on some attribute, with one defined as internal and the other as external, the E-I Index is

$$EI = EL - IL / EL + IL$$

where EL equals the number of external ties and IL equals the number of internal ties. EL was defined as the number of persons outside the focal person's assigned care team with which the focal person interacted based on the sensors, and IL was defined as the number of persons outside the focal person's assigned team with which the focal person

interacted. An EI value of +1.0 means all ties were outside the team (high boundary spanning), 0 means that ties were divided equally, and -1.0 means that all ties were within the team (low boundary spanning).

Covariates. We included gender, another individual characteristic, as a covariate because research suggests that females generate more creative ideas (George, 2007). We also included the following center characteristics to capture differences in patient profile and workload: percentage of patients uninsured and number of patient visits per full-time employee during 6-month interval, because such operational factors may shape the context for creativity. We included a center-level indicator of organizational and coworker support—the cultural factor psychological safety, which refers to the shared belief that the setting is safe for interpersonal risk taking (Edmondson, 1999), a belief previously found to influence employee creativity (Kessel, Kratzer, & Schultz, 2012). We measured psychological safety using center means of staff survey responses to three validated items, such as “Members of this team are able to bring up problems and tough issues” (Edmondson, 1999). Inclusion of this factor is also important given Zhou & George’s (2001) finding that organizational and coworkers’ support influences creativity of dissatisfied workers. Finally, a time dummy variable was included to account for the month (out of 18 months) during which each idea was generated.

Statistical Analyses

We first calculated bivariate correlations to assess the unadjusted associations among all variables and multicollinearity. We then performed a series of multilevel regression models at the level of the idea ($N = 220$ ideas) to estimate the associations between job dissatisfaction, person–job characteristics, covariates, and idea creativity. We used the SAS GLIMMIX procedure to account for the multilevel structure of the data (ideas nested in individuals, located in centers), multiple creative ideas per individual, and a continuous outcome variable. We estimated standard errors accounting for clustering at the individual and center-level and calculated variance inflation factors as an additional check for multicollinearity.

In the first model, we assessed the association between job dissatisfaction and idea creativity, adjusting for covariates to examine whether there is a main effect on average. In the second model, we tested our remaining hypotheses by adding our proposed moderators and their

interactions with job dissatisfaction to our first model. To reduce multicollinearity between interaction terms and their components, we standardized our variables before entering them in the models. We examined the standardized coefficients produced by the models and their p values to evaluate the significance of effects and thus support for our hypotheses. When an interaction term was significant, we calculated the effect of job dissatisfaction on idea creativity for individuals who were high versus low on the relevant moderator (i.e., central vs. supporting role or one standard deviation above vs. below mean otherwise) and then plotted the simple slopes for each group in figures to illustrate the effects.

Results

Table 1 presents descriptive statistics and bivariate correlations for all study variables. The table shows that the average creativity of the 220 improvement ideas offered by individuals was 2.47 out of 5 ($SD = 1.64$) and job dissatisfaction for the average individual was 3.46 out of 4 ($SD = 0.64$). Person–job characteristics were well-distributed, although the mean level of boundary spanning was higher than might be expected ($M = 0.64$) though consistent with recent national efforts to encourage such. Correlations between person–job characteristics were significant (p values $< .05$); however, variance inflation factors were all less than 6 (under the threshold of 10), indicating that multicollinearity was not a major concern.

Table 2 presents the results of the multilevel analyses used to test our hypotheses. Model 1 shows that the emotional experience of job dissatisfaction was positively associated with idea creativity ($p < .0001$), supporting Hypothesis 1. A 1-standard deviation increase in job dissatisfaction was associated with a 0.32-standard deviation increase in idea creativity on average. Model 2 shows that the interactions of our focal person–job characteristics—role centrality, organizational tenure, and boundary spanning—with job dissatisfaction were significant with respect to idea creativity as hypothesized (p values $< .05$). The Akaike information criterion goodness-of-fit statistic decreased in magnitude with the inclusion of the interaction terms in the model, suggesting a better fit was associated with this model (vs. the main-effects models). The nature of the interactions is depicted in Figure 2.

Figure 2(a) indicates dissatisfaction had a positive, significant association with creativity for individuals with shorter organizational tenure ($\beta = .39$, $p = .01$) and a positive but insignificant association for those

Table 1. Descriptive Statistics and Correlations of Key Variables in Analyses (N = 220 Ideas Generated by 72 Staff).

	M	or %	SD	1	2	3	4	5	6	7	8
1. Idea creativity (1–5)	2.51		1.65								
2. Job dissatisfaction (1–4)	3.46		0.64	0.39**							
3. Organizational tenure (1–6)	3.07		1.49	–0.50**	–0.19*						
4. Role centrality (1 = central; %)	0.62		0.45	–0.52**	–0.24*	0.37**					
5. Boundary spanning (0–1)	0.64		0.31	0.08*	–0.20*	–0.04	0.14**				
6. Gender (1 = male; %)	0.20		0.40	0.06	–0.27**	–0.05	0.12	–0.17			
7. Patients uninsured (%)	0.07		0.05	–0.08	0.01	0.09	–0.05	0.05	–0.09		
8. Number of patient visits per staff	417.36		81.55	0.05	–0.13*	0.05	–0.10	0.05	–0.30	0.17	
9. Psychological safety (1–4)	2.83		0.75	0.56**	0.27*	0.32*	–0.18	0.21	–0.32	0.06	–0.22*

* $p < .05$. ** $p < .001$.

Table 2. Results Analyses Predicting Idea Creativity (N = 220 Ideas).

	Dependent variable: idea creativity	
	Model 1	Model 2
Independent variables	β (SE)	β (SE)
Job dissatisfaction	.32 (0.04)**	.46 (0.13)**
Organizational tenure		-.03 (0.02)**
Organizational Tenure $\times$ Dissatisfaction		-.07 (0.01)**
Role centrality		.23 (0.40)
Role Centrality $\times$ Dissatisfaction		.27 (0.10)**
Boundary spanning		.01 (0.19)
Boundary Spanning $\times$ Dissatisfaction		.05 (0.02)**
Covariate—individual-level		
Gender (1 = male)	-.27 (0.16)	-.13 (0.15)
Covariates—center-level		
Percentage of patients uninsured	-.59 (0.79)	-.28 (0.81)
Number of patient visits per staff	.07 (0.01)	.06 (0.01)
Psychological safety	.81 (0.07)**	.86 (0.06)**
Time (month idea raised)	Yes	Yes
Center	Yes	Yes
AIC (smaller is better)	420.33	390.22

Note. β is standardized regression coefficient; SE is standard error. AIC = Akaike information criterion.
** $p < .001$.

with longer organizational tenure ($\beta = .11$ $p = .61$). Figure 2(b) shows dissatisfaction was positively associated with creativity for individuals with central roles ($\beta = .46$, $p = .001$), whereas it had a positive yet not significant effect on creativity for those in supporting roles ($\beta = .21$, $p = .35$). Figure 2(c) shows dissatisfaction had a positive relationship with creativity for individuals high on boundary spanning ($\beta = .37$, $p = .01$) and a positive and marginally significant effect on creativity for those low on boundary spanning ($\beta = .32$, $p = .09$). These results suggest that there is a positive association between job dissatisfaction and creativity for individuals with a shorter organizational tenure, more central roles in the care team (here, PCPs and nurses), and high boundary spanning consistent with Hypotheses 2, 3, and 4. For those with supporting roles, longer tenure, and low boundary spanning, level of dissatisfaction had limited effect on their creativity, which tended to be lower than their counterparts' on average. Additional analyses with the

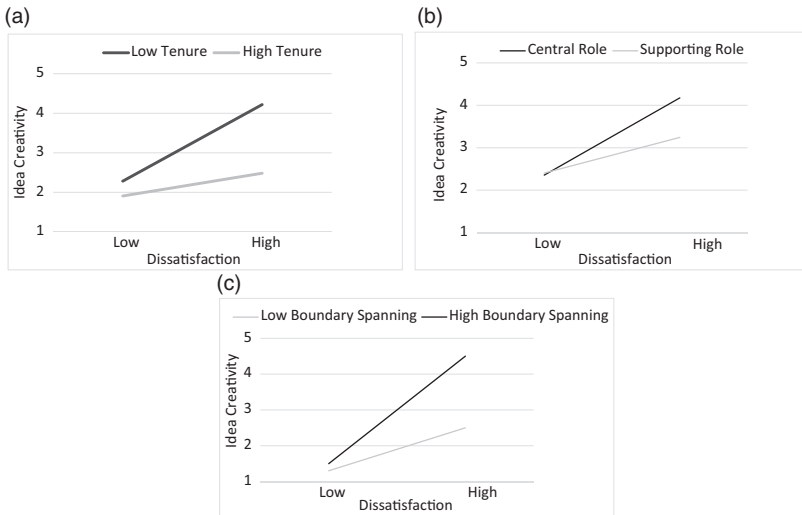


Figure 2. Impact of moderating variables on the job dissatisfaction and idea creativity relationship: (a) organizational tenure as moderator, (b) role centrality as moderator, and (c) boundary spanning as moderator.

squared terms for dissatisfaction and person–job characteristics did not yield significant results, limiting the ability to identify optimal levels of these variables for creativity.

Psychological safety was the only statistically significant covariate across all models (p values < .001). Consistent with prior research (Kessel et al., 2012), a greater sense of psychological safety was associated with greater idea creativity.

Discussion

These findings highlight that dissatisfaction, though a negative emotion associated with withdrawal and exit in many settings, need not lead to such negative events for workers and organizations. Dissatisfaction can be associated with positive processes that benefit individuals, clients, and ultimately organizations—under certain conditions. Our results indicate that dissatisfaction is positively associated with creativity under the extreme work conditions of health care delivery. We hypothesized this association based on theory that dissatisfaction provides cognitive persistence (a pathway to creativity) in contexts like health care, where extreme conditions are activating. While an

organization would certainly not want to foster dissatisfaction, our findings suggest that dissatisfied creators exist, that is, individuals who generate positive ideas for improvement amid negative emotion. Thus, there exists a paradoxical relationship of negative emotion linked to positive cognitive process in extreme work settings.

In addition, our findings indicate person–job characteristics moderate the dissatisfaction–creativity relationship in the extreme work setting of health care. Consistent with Hypotheses 2, 3, and 4, tenure, role centrality, and boundary spanning each altered the relationship. Shorter tenure, role centrality, and boundary spanning—which increase knowledge search, knowledge depth, and knowledge breadth, respectively—strengthened the effect of dissatisfaction on creativity. Individuals with greater dissatisfaction and each or more of these characteristics generated ideas with greater creativity. Each characteristic arguably enhances cognitive flexibility—finding and processing divergent information—a key ingredient for creativity.

Altogether, our findings support our assertions that the dissatisfaction–creativity relationship is contingent on two aspects of work: the nature of work, specifically, the extremity of work, and person–job characteristics. As such, our results suggest two additional categories of moderators affect this relationship, and potentially the relationship between other negative emotions and other positive cognitive processes related to work more generally. In addition, as the nature of work may drive cognitive persistence and person–job characteristics influences cognitive flexibility, our results suggest that both cognitive pathways to creativity are operable in health care and potentially other extreme work environments.

Conclusion

Our results identifying dissatisfied creators and person–job characteristics that further accentuate their creativity extends past research on negative emotion and creativity and offers avenues for future research. First, our finding that extreme work is a context that yields dissatisfied creators extends prior work identifying contextual support as a moderator and suggests value in future research continuing to identify contexts where these individuals exist, whose ideas might be leveraged to improve work. Prior work by Zhou and George (2001) identified coworker and organizational support as moderators of the relationship between dissatisfaction and creativity, highlighting the importance of social context. Our study shows that beyond a psychologically safe, supportive context, extreme work contexts are another condition that supports dissatisfied creators and thus highlights the importance of the

industry and nature of work as well. Future work should assess whether other settings prompt similar dynamics.

Future research should also consider other, noncontextual moderators of emotion–creativity relationships. Past work found direct effects of person–job characteristics similar to the constructs studied in this work (e.g., being close to the customer, boundary spanning orientation) on creativity and innovation (Kanter, 1988). We found these characteristics to be significant moderators as well. This finding indicates there can be greater precision in identifying individuals who are more likely to activate when dissatisfied rather than exit or remain passively with the organization. Past work found continuance commitment, coworker, and organizational support variables to moderate the emotion–creativity relationship (Zhou & George, 2001). Continuing to identify variable classes that moderate dissatisfaction and other emotions can add precision to our understanding of the antecedents of creativity. While these variables may not all be malleable, they help clarify how creativity is the result of context and individual.

As dissatisfaction is a nontransient emotional state (compared with transient moods), future research could reveal if other nontransient emotional states also are associated with creativity through the mechanisms suggested by the DPCM. We theorize that dissatisfaction prompts cognitive persistence to activate creativity, extending the DPCM beyond transient mood states to nontransient emotional states (De Dreu et al., 2008). Other nontransient emotional states that could be of interest to researchers and practitioners include engagement, thriving, affective organizational commitment, and affective well-being (Fisher, 2010). Our findings suggest that more research is needed to understand if nontransient emotional states generally follow the pathways to creativity shown by the more temporally bound transient mood states.

In this study, we built on the DPCM to theorize that dissatisfaction influences creativity via cognitive persistence. It may be that dissatisfaction evokes both persistence and flexibility. Another possible mechanism for this relationship is ambivalence, which is the simultaneous experience of positive and negative emotion (Fong, 2006). Individuals experiencing emotional ambivalence are better at recognizing unusual relationships between concepts, which may have relevance to generating creative ideas for improvement (Methot, Melwani, & Rothman, 2017). Extreme and unique work settings such as health care combined with an individual's emotional ambivalence may also increase the likelihood of idea creativity. Future research should empirically test mechanisms by which dissatisfaction relates to creativity simultaneously to provide

insight into the relative contribution of mechanisms and a more comprehensive model of emotion-linked creativity.

Future research should also seek to address methodological limitations of this study. Our study design does not allow causal inference. Longitudinal examination of dissatisfaction and creativity is needed to uncover if dissatisfaction predicts creativity, occurs simultaneously with it, or results from creativity. It is possible that the association that we found operates in a different direction: Those who generate creative ideas may become dissatisfied with the status quo. In addition, we were unable to include creativity-focused constructs such as creative traits or thinking styles, which are common individual-level factors tested in creativity research. Third, we used data from a small sample of community health centers. The findings may not generalize to other types of health care organizations nor to all primary care clinics because of differences between the focal clinics and other primary care clinics. Further, our sample of quality improvement team participants were not a random sample as they may have been a particularly dissatisfied group motivated to participate in the team to improve current services or practice. Finally, all creative ideas in this study were voiced in meetings. We do not capture creative ideas that were generated but not voiced or voiced elsewhere, a limitation although voicing is necessary for improvement and these meetings were a key forum.

Our results suggest that managers should leverage worker job dissatisfaction in health care delivery as an opportunity to generate creative ideas. It may appear that few design solutions exist to foster creativity from dissatisfied creators, as the person–job characteristics we highlight may not seem modifiable. However, organizations can launch initiatives that account for these characteristics. For example, organizations can target elicitation efforts to workers who have creative potential. Soliciting their participation may require engaging these individuals directly and providing opportunities for them to contribute that allow their feedback to be carefully considered (e.g., innovation tournaments, improvement team participation, or online work-in-progress sessions; Schwartz, 2018). The high dissatisfaction of improvement team members in this study suggests that the dissatisfied can be willing to be engaged. Another solution is to select dissatisfied creators as participants in “care co-design initiatives” with leaders, patients and families, and workers (Institute of Medicine, 2012). To further bolster their creativity, dissatisfied creators could be encouraged to collaborate on idea generation. For example, problems requiring complex patient-related knowledge may benefit from ideas of dissatisfied creators in

central roles, and these ideas may be enriched by the fresh perspective of newer tenured dissatisfied creators. Experimenting with opportunities for cross learning and peer mentoring between dissatisfied creators, and other workers, could yield greater creativity. Research on creative freelancers shows that pairing newcomers with experienced workers in teams provides resources such as technical feedback that are important to ensuring the success of creative ideas in the specific organizational environment (Schwartz, 2018).

This study shows merits of taking a closer look at dissatisfied creators and positive outcomes like creativity that may arise from negative emotions like job dissatisfaction. Attending to the dissatisfied may not only improve workers' well-being but also benefit customers (patients) and organizations. Dissatisfied creators, particularly those with certain person-job characteristics, can be valuable resources for creative improvement ideas in health care and potentially other extreme work settings.

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